

Received

Your poster submission has been received

Print Print this page now.

You have submitted the following poster to Plant and Animal Genome Conference / PAG 34

(January 8-12, 2027). Receipt of this notice does not guarantee that your submission was complete, free of errors, or accepted for presentation.

An Integrative Bioinformatics Framework for Transcriptomic Reanalysis Reveals Regulatory Architectures In the Lychee Response to *Peronophythora*

Eric Zhuang, Great Neck South High School, Great Neck, NY

Abstract Text:

Lychee (*Litchi chinensis* Sonn.) is an economically important tropical fruit threatened by litchi downy blight, caused by the oomycete *Peronophythora litchii*. To prioritize genes and regulatory features associated with early host responses, we reanalyzed public leaf RNA-seq data from Guiwei and Yurong cultivars (GSE201243; mock and inoculated samples, three biological replicates per group). Differential-expression analysis identified 17 significant DEGs in Guiwei and 117 in Yurong at adjusted $P < 0.05$, suggesting a broader transcriptional response in Yurong. Eighteen highly responsive genes were selected for structural, motif, promoter, and candidate interaction analyses. These candidates included annotations associated with defense or stress responses, including jacalin-like lectin, pathogenesis-related protein, heat-shock protein, thaumatin/osmotin-like proteins, cytochrome P450, lipid-transfer proteins, and calcium-binding proteins. A selected-gene interaction contrast prioritized LITCHI019519, LITCHI001510, LITCHI028401, LITCHI017676, LITCHI028104, and LITCHI019183 as potential cultivar-dependent response genes. Promoter analysis of 2-kb upstream regions detected stress- and hormone-associated cis-elements, and a random-promoter background test supported enrichment of ARE, MeJA-responsive motifs, TCA-elements, and TC-rich repeats. These results provide testable candidates for future functional validation of lychee resistance to *P. litchii*.

Session Selection:

Fruit Species

Final Number:

Title:

An Integrative Bioinformatics Framework for Transcriptomic Reanalysis Reveals Regulatory Architectures In the Lychee Response to *Peronophythora*

Submitter's E-mail Address:

ericzhuang2010@gmail.com

First Presenting Author

Presenting Author

Eric Zhuang
ericzhuang2010@gmail.com

Student Researcher
Great Neck South High School
341 Lakeville Road
Great Neck, NY 11021
USA

If necessary, you can make changes to your poster submission between now and the deadline of

Thursday, December 31, 2026

To access your submission in the future, use the direct link to your poster submission from one of the automatic confirmation emails that were sent to you during the submission.

Or point your browser to <http://pag.confex.com/pag/reminder.cgi> to have that URL mailed to you again. Your username/password are 64365/760141.

Any changes that you make will be reflected instantly in what is seen by the reviewers. You DO NOT need to go through all of the submission steps in order to change one thing. If you want to change the title, for example, just click "Title" in the poster control panel and submit the new title.

When you have completed your submission, you may close this browser window.

Tell us what you think of the poster submittal

[Home Page](#)